

Predicting Species-Infrastructure Collision Risks Across Different Environments: Spatial Analysis Using Machine Learning

Agung Wahyu Brian Mahandes Putra Blackbeard
Sekolah Dyatmika Bali, brianblackbeard08@gmail.com

Abstract

Collisions between wildlife and human infrastructure are a growing global concern, threatening biodiversity and human safety. This study focuses on predicting two major types of wildlife-infrastructure collisions: Wildlife-vehicle collisions (WVCs) on roads and bird collisions with wind turbines. In this paper, I will examine how various machine learning models can be utilised to predict collisions across different environments by integrating spatial, temporal, and biological data. Using three datasets across two different environments, we identify patterns between animal/bird species, time of year, land cover types and geological factors to determine which species is at most risk of collision given a certain scenario. These machine learning techniques will also help identify which of those factors plays the biggest role in these collisions. The findings of this research could support wildlife conservation efforts and inform infrastructure planning to reduce roadkill and bird turbine collision incidents, such as wildlife bridges, road signage, and targeted policy interventions during high-risk seasons. Overall, Random Forest Classifiers and other ensemble learning methods have proven to work the best across different environments. With Random Forest in particular, achieving the highest F1 score in each dataset used - anywhere from 1-16% higher than the next best model.

1. Introduction

The expansion of human infrastructure has created new and often deadly hazards for wildlife. As roads, highways, and energy systems stretch across the landscape of our globe, animals are increasingly exposed to collision risks with our manmade infrastructure. Two particular types of these species-infrastructure interactions are Wildlife-vehicle collisions (WVCs) and bird strikes on wind turbines. These incidents not only contribute to the decline of animal populations but can also threaten human safety.

Wildlife-vehicle collisions (WVCs) are an increasingly urgent issue around the world. As roads and highways expand into natural habitats, animals can be forced to cross these human infrastructures to find food, mates or shelter. These crossings can very often result in injury or death for the animals if they are encountered by an oncoming vehicle. As a result, these collisions can disrupt ecosystems and contribute to a loss in biodiversity, which is especially true for species already at risk. It's estimated that anywhere from one to two million vertebrates die each year on roads in the US alone [1].

Moreover, while conservation efforts are already an adequate reason as to why protective measures should be put in place to reduce these WVCs, human safety and infrastructure are also at risk in these interactions, further pushing the need for further research to be done on the topic.

The risks of wildlife-vehicle collisions vary widely depending on the time of year, the terrain, the type of environment, the species involved, and more. For instance, animals may be more likely to be on the move during their breeding seasons or when migrating, making them more vulnerable to collisions with human infrastructure. The habitat surrounding the roads also plays a key role; forests, grasslands, and wetlands each present different movement patterns and therefore pose different risks [2].

Moreover, in addition to these natural features, the volume, average speed, and other characteristics related to vehicles moving across these roadways will also play a pivotal role in determining which species are at most risk and how much at risk those said species are [3].

To reduce the number of collisions and protect both wildlife and drivers, researchers have started using predictive machine learning modelling techniques. These include models that can detect patterns in data and predict which species is likely to be at most risk of collision, considering the geographical and spatial features of a certain location. Using machine learning approaches to analyse large, complex datasets with many unique variables can produce accurate predictions/results. These predictions can then inform the design of safer infrastructure, like wildlife bridges, fencing, or speed limits in high-risk areas.

Moreover, these models can also work to calculate which geographical or biological features of different species and terrain play the most pivotal role in promoting a Wildlife Vehicle Collision (WVCs). Finding what contributes most to a collision is a key component in preventing more from occurring, and this can be difficult to do as collision risk can vary significantly based on species behaviour, habitat type, season, and local climate. Likewise, environmental factors like land cover type or topography can influence how animals move and interact with human infrastructure.

In addition to roadkill, this paper will also investigate collisions between birds and wind turbines. With the growth of wind energy, which is essential in reducing our greenhouse gas emissions, wind turbine farms have grown exponentially worldwide [4]. Whilst necessary as we move to a greener and renewable future, such infrastructure has introduced new challenges for bird conservation. Many species of birds, especially those of migratory status, are vulnerable to turbine collisions, particularly in cases where such turbines are placed along migratory routes or near nesting sites.

In this project, I will explore how different machine learning models can be used to predict wildlife-infrastructure collisions across different environments, namely: WVCs and bird-turbine collisions. By combining spatial (location, geographical features, land cover, etc), temporal (month, season), and biological (species-level) data, this paper not only aims to predict these collisions, but also aims to find which features are most pivotal in causing these collisions. I have used three real-world datasets for this analysis; their details are listed in 3.1.

Our end goal is to contribute to the growing field of conservation technology by applying data-driven machine learning methods to a real-world problem. This research can support both wildlife conservation and the development of safer, more sustainable infrastructure.

2. Literature Review

Predicting Wildlife Vehicle Collisions is a critical area of research within the conservation element of technology. WVCs are increasingly becoming a major contributor to biodiversity loss [5], especially among avian and terrestrial species that regularly interact with human infrastructures: Roads, Highways, Wind Turbines, etc. Machine learning models present a promising tool in supporting large-scale ecological forecasting when combined with large, labelled datasets. Allowing us to reduce the risk of these collisions.

A recent study integrating deep learning techniques to forecast Avian Migration Patterns was conducted by Kehinde Owwoye, 2020 [6]. Owwoye proposed a deep bidirectional RNN to forecast hourly migration states of Turkey vultures using a combination of different topological data. By aiming to forecast the longitude coordinates with the most informative features, the method outperformed classical baselines like Logistic Regression, Ada Boost, Support Vector Machine and Dynamic Neural Networks. Demonstrating that the more complex Stacked Bidirectional GRU (Bi-GRU) are more effective than traditional regression techniques in capturing the nuanced dynamics of avian migration across continents.

The study also presented that environmental variables such as Population density, Albedo and Elevation were shown to have high correlation with the timing of migration transitions. Building on previous ecological insights, whilst offering new perspectives on the more pressing predictive indicators, as factors such as

Sunshine Duration and Evaporation had lower correlation with the migration transitions. Overall, these findings support the broader argument that data-driven ecological forecasting can meaningfully inform infrastructure management, wildlife policy, and hopefully lead to lower biodiversity loss. Furthermore, it benefits conservationists and transportation planners in working dynamically in environments with a high risk of WVCs.

3. Methodology

3.1. Dataset Description

Dataset	Description
Eagle_subset.csv	Dataset consisting of 10000 data points. 12 input features and 1 output. Those 12 features describe both the conditions of the terrain/climate, and the final output column describes whether or not the bird collided with a wind Turbine. Already pre-split into train (66.3%) and test (33.7%). Classification Task.
newvulturedata.csv	Dataset consisting of 179989 data points and 40 input features. Later filtered to 28621 data points with 25 input features and 1 output column. 25 features described Split into train and test sets in an 80/20 ratio. Forecast & Classification Task.
Traffic_roadkill_data_2023.xlsx	Dataset consisting of 18745 data points and 123 input features with 1 output column. Later filtered to 8607 data points - after removing empty/missing data. 123 input features describe the topography of the area alongside the climate and season at the time of collision. The output column describes the species that was involved in the incident. Split into train and test sets in an 80/20 ratio Classification Task.

All machine learning models (Excluding RNNs and ANNs) were trained using the Scikit-learn library on Jupyter Notebook on Apple M4

ANNs and RNNs were trained using the Keras deep learning API in TensorFlow on Google Colab using an Nvidia T4 GPU.

3.2. Data Preprocessing

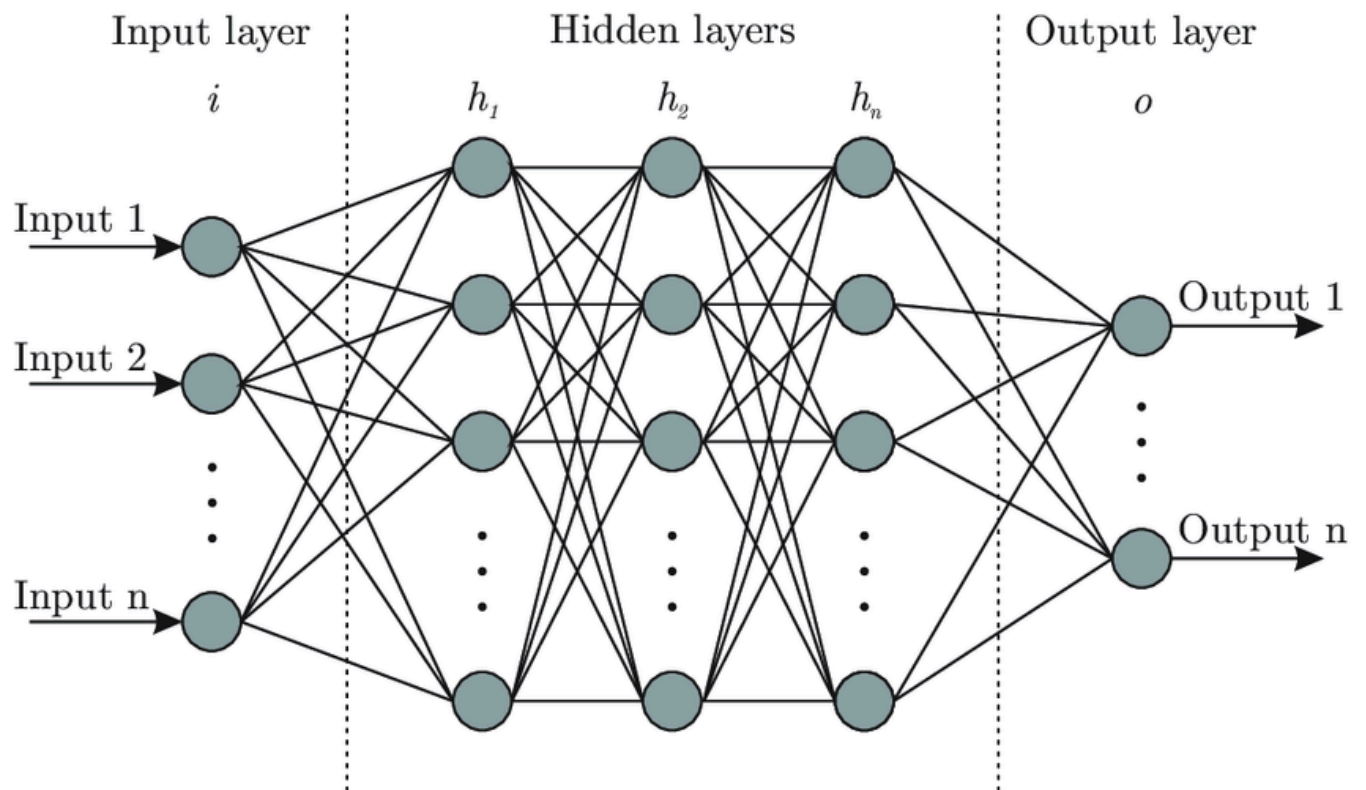
For all three datasets, I applied relevant Data Preprocessing steps to improve model learning and obtain greater performance scores. For all categorical data in the datasets, I applied **one-hot encoding** to transform it into numerical formats. Allowing all the machine learning models to be utilised to understand the data effectively. Furthermore, to allow usage of logistic regression, I applied scikit-learn's **standard scaler** to all the numerical data in our datasets. Finally, I implemented a **Simple Imputer** to fill in any missing values in our datasets. For this specific imputer, I used the mean value of the specific feature, and I found this to be especially needed in both the Eagle Subset Dataset and the Roadkill Dataset due to their small size. If I were to instead remove data with missing values, it could've potentially led to less data to train the models, which is already an issue with those datasets, as they already lack data.

3.3. Models

In this study, I chose from a wide pool of both neural network architectures and supervised machine learning algorithms which suit this classification problem at hand. Specifically, I'll describe the eight models (ANN, RNN, Logistic Regression, KNN, Random Forest, Support Vector Classifier, Gradient Boosting, AdaBoost) that I have used for this project.

1. Simple ANN (Artificial Neural Network)

Artificial Neural Networks are central models in the large field of deep learning. These complex systems - which are highly capable of capturing non-linear relationships in data - are made up entirely of a multitude of different layers, layers which in turn are constituted of many individual nodes. Each node has a global input - built up from its connections with each of the nodes from the prior layer - which is then transformed through a function into an output value, that is passed onto every node following. This dense formation of fully connected layers makes ANNs particularly effective & highly robust in classification problems, allowing them to make predictions from large inputs of data [7].



In a standard ANN architecture, the initial layer is an **input layer**, whereby all the raw feature vectors are inputted before being processed through multiple **hidden layers**. These layers obtain increasingly fine details/representations of the input by slightly adjusting the weights and biases stored on each node, eventually enabling the network to learn complex decision boundaries [7]. Following this is the **output layer**, a layer which consists of several nodes that correlate to the number of different predictions which can be made (e.g in a problem where the results could be either male or female, the output layer would have 2 nodes).

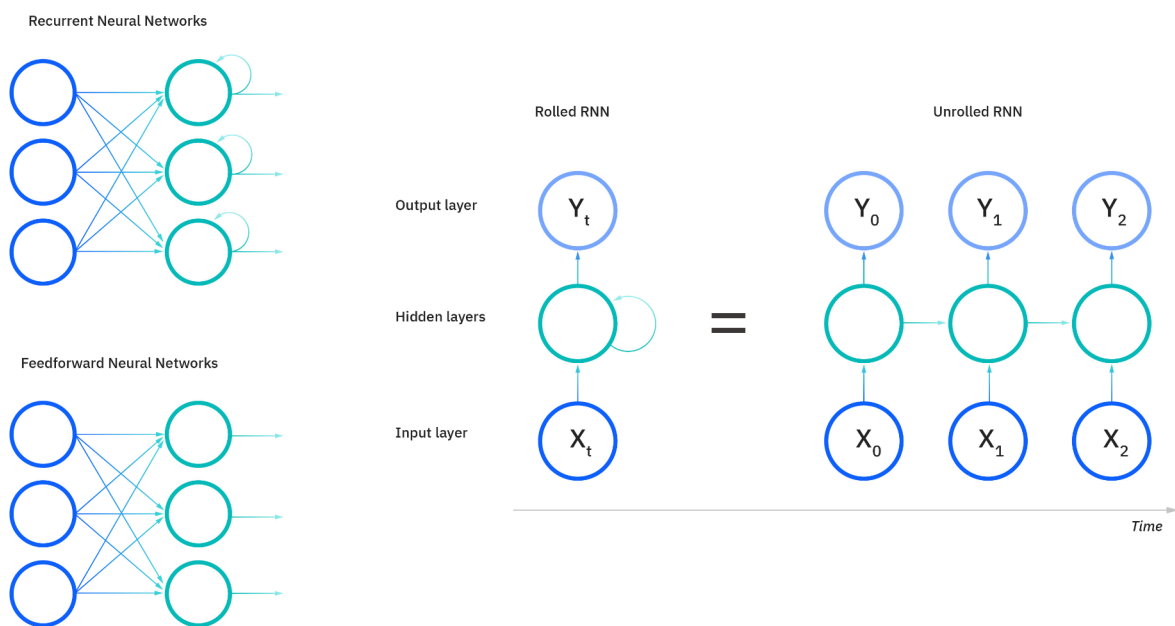
In my specific implementation (for all 3 datasets), I used 2 hidden layers, each with 256 nodes, to encourage feature generalisation. Both layers used the ReLU (Rectified Linear Unit) activation function and were both followed by a BatchNormalization layer. A layer used to standardise activations and smoothen gradient flow during training, which in turn promotes more accurate results [8]. My final output layer consisted of either 3 or 4 nodes (depending on the dataset) and used the softmax activation function. The model is trained using the Adam optimiser with a sparse categorical cross-entropy loss function, highly suitable for multi-class classification tasks.

Furthermore, I also applied an **early stopping function** to prevent overtraining of the neural network and restore the weights & biases of the model to its best results.

Early Stopping Function: A Function that stops training of the neural network if the model's test dataset's accuracy begins to fall as opposed to rise with more epochs. This is due to the model overfitting the training dataset.

2. RNN (Recurrent Neural Network)

Similar to ANNs, RNNs are constituted of Input, Hidden and output layers, however, with the additional feature that they can 'retain' memory from prior inputs to assist with decision-making for further predictions. To do this, RNNs have special loop-like structures that provide context to influence current processes, making them particularly powerful in situations where the order of occurrences has an effect.



As seen from the diagram above, each RNN unit takes both an input and the previous unit's **hidden state**, and outputs its hidden state as a result. These hidden states are essentially a 'memory' of what occurred beforehand in the sequence [9].

In my specific implementation (for all 3 datasets), I used 1 RNN layer, with 256 nodes to encourage feature generalisation. For this layer, I used the Tanh activation function and followed it with a BatchNormalization layer. I then used the softmax activation function with my output layer.

The model is trained using the Adam optimiser with a sparse categorical cross-entropy loss function as the ANN above. As well as using the same early stopping function.

3. Multinomial Logistic Regression

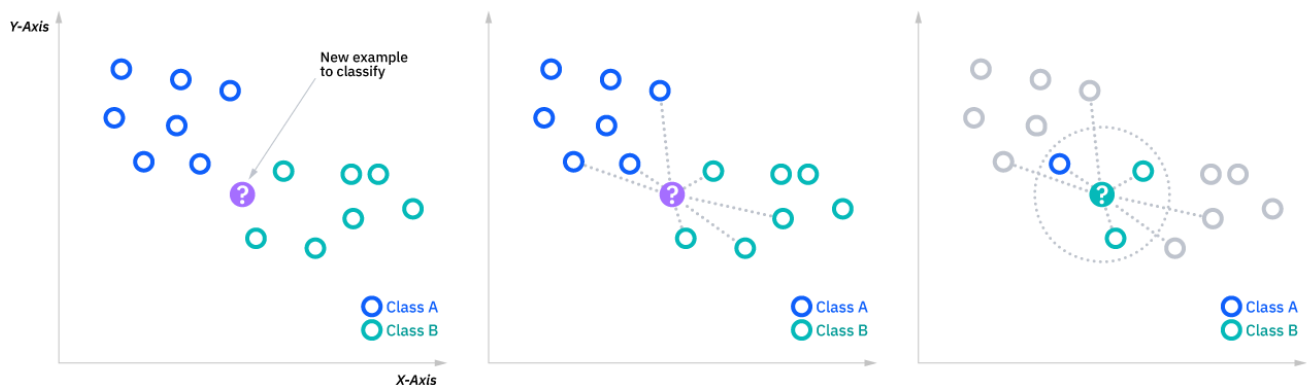
A fundamental statistical method, Multinomial logistic regression is a model which allows us to predict which category a subject will fall under, given its input variables. Similar to Artificial Neural Networks, the model begins with an initial **input layer** of nodes, whereby all the raw feature vectors are inputted. These nodes are all fitted with their vectors of weights and biases, depending on how significant a role they play in deciding which category a subject will fall into. As this is a supervised machine learning model, these weights and biases will be adjusted during training to optimise the algorithm [10].

However, unlike Artificial Neural Networks, Multinomial Logistic Regression models do not have **hidden layers**, and instead directly feed into the **output layer**. This simplicity makes the MLR an especially powerful baseline classifier for evaluating more complex models (such as ANNs).

4. KNN (K-Nearest Neighbours)

A simple and popular regression classifier, KNNs (K-Nearest Neighbours) are supervised learning models which work on the assumption/principle that like outcomes & predictions could be found near one another.

Unlike traditional ML models, which learn from a training dataset, KNNs simply base their predictions on their training dataset. The model, using this pre-existing training data, will calculate the **closest** “K-neighbours” from a new data point. Following this, the predicted value of this new data point is then formulated based on which value is most frequently occurring in these “K-nearest neighbours” [11].



An example illustrated above: Here we are predicting whether a subject falls into class A or class B, considering two inputs (Y and X). As seen from the third diagram, the K value in this KNN is 3 (the 3 nearest neighbours). As the majority of those 'neighbours' are class B, we can then predict that this new subject is class B.

5. Random Forest Classifier

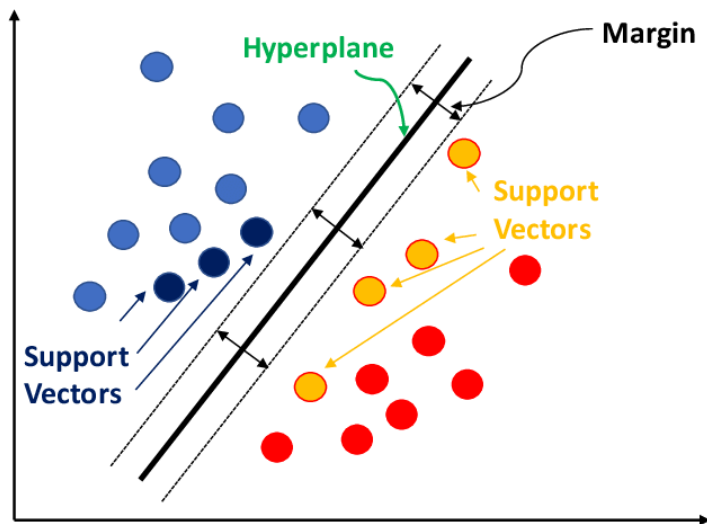
Another fundamental Machine learning model, Random Forrest Classifiers are algorithms built up of several smaller **decision trees**. A single decision tree makes a prediction based on a subset of the training dataset, so by aggregating the predictions of multiple decision trees - a process known as **ensemble learning** - the model is able to make a more generalised prediction less prone to **overfitting**.

A key feature of Random Forest classifiers is their usage of Bootstrap Aggregation. An ensemble learning method, Bootstrap Aggregation, is an algorithm that randomly samples data from the training dataset for each **decision tree** to learn from. These samples can differ or coincide with individual data points, but overall will promote a model less prone to overfitting and therefore increase the accuracy of its predictions [12].

6. Support Vector Classifier

Support Vector classifiers - also known as Support Vector machines - are machine learning models which aim to find a decision boundary/hyperplane that best separates different classes within the training dataset. To do this, these algorithms work to maximise the margin - the distance between the hyperplane and the nearest data point from each class. The planes running parallel to this 'hyperplane'/decision boundary, which include those nearest data points, are known as **support**

vectors. By focusing solely on these support vectors, the model can build robust decision functions which generalise well to unseen data.



Another reason why Support Vector classifiers are commonly applied in the world of data science is due to their ability to handle both linearly and non-linearly separable data. This is accomplished via the help of kernel algorithms - algorithms which implicitly map the input features into a higher-dimensional space and therefore allow the model to

learn more complex decision boundaries. Further improving the overall model performance and accuracy [13].

7. Gradient Boosting

Similar to Random Forest classifiers, Gradient boosting combines several decision trees into a single model to make predictions. Another form of **ensemble learning**, gradient trees, however, differ from random forest in that each decision tree is trained additively as opposed to in parallel and independently. Each new tree attempts to correct the mistakes of the last; therefore, eventually, by focusing on the weaker areas of the model and incrementally adding new steps, each new prediction should gradually grow more correct.

By progressively refining a single unified model, gradient boosting algorithms tend to be more powerful and accurate than their similar counterpart - random forest models. However, without the many diverse trees of random forest, which aid in reducing its variance, gradient boosting algorithms are more prone to **overfitting** the training dataset [14].

8. Ada Boost

Ada boost - short for adaptive boosting - is an ensemble learning method very similar to gradient boosting. The model works by sequentially training several weak classifiers - known as decision stumps - by giving more weight and bias to misclassified samples after each iteration of the training batch. Therefore, this causes the model to focus harder on misclassified/harder cases. Once the model is sufficiently trained, the final prediction is made through a weighted vote of all the 'learners'/classifiers, with stronger and more accurate learners receiving a higher influence in the vote. This statistical technique tends to be effective at reducing bias and variance; however, it can be highly susceptible to noisy data and outliers since misclassified points receive much higher attention.

3.4. Metrics

The metric I used to evaluate the performance of both the different neural network architectures and the several machine learning models is the F1 score. A fundamental method to evaluate both binary and multiclass classification problems - the latter of which is what this paper is researching here - the F1 score is the harmonic mean of both the **precision** and **recall** of a classification model. Generally, due to it taking into account the nature of the errors (True Positive, False negative, etc), it is a more useful and superior evaluation metric than plain **accuracy** especially in the case of imbalanced datasets (share of certain outcome/class in dataset has a much larger share than other outcomes (i.e 80/20 split)).

$$F_1 = \frac{2}{\frac{1}{Recall} + \frac{1}{Precision}} = 2 \times \frac{(Precision \times Recall) \times 1}{(Precision \times Recall) \times \frac{Precision + Recall}{Precision \times Recall}} = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

As the F1 score is a fraction, a score nearing 1 is indicative of a good result (both precision and recall were nearing 1), and therefore, a score nearing 0 is indicative of a poor result - the model was mostly incorrect with its predictions.

Accuracy: $(TP + TN) / (TP + FP + TN + FN)$ Overall correctness

Precision: $TP / (TP + FP)$. It refers to the proportion of positive predictions which were truly positive

Recall: $TP / (TP + FN)$ refers to the proportion of all actual positives which were correctly classified as positive

4. Experiments and Results

4.1. Machine Learning Models

For each dataset, I applied one-hot encoding to categorical features and used Scikit-Learn's standard scaler for every numerical feature to allow the model to work with logistic regression. Moreover, every dataset was split into an 80/20 ratio for training and testing sets.

Eagle Subset

Model	Test Accuracy	F1 Score
ANN	60.68%	41.11%
RNN	63.59%	27.24%
Logistic Regression	62.43%	37.63%
KNN	59.88%	41.95%
Random Forest Classifier	64.87%	42.91%
Support Vector Classifier	64.84%	40.30%
Gradient Boosting	64.93%	42.18%
Ada Boost	61.45%	39.51%

New Vulture Data

Model	Test Accuracy	F1 Score
ANN	95.79%	86.84%
RNN	99.82%	49.96%
Logistic Regression	94.06%	80.68%
KNN	98.13%	95.07%
Random Forest Classifier	99.60%	99.11%
Support Vector Classifier	97.83%	93.97%
Gradient Boosting	99.39%	98.47%
Ada Boost	96.61%	91.77%

Traffic Roadkill Data

Model	Test Accuracy	F1 Score
ANN	52.47%	39.37%
RNN	44.47%	27.54%
Logistic Regression	23.17%	21.86%
KNN	40.36%	25.32%
Random Forest Classifier	61.36%	55.87%

Support Vector Classifier	33.41%	32.40%
Gradient Boosting	48.88%	37.36%
Ada Boost	34.98%	05.76%

From the three tables above, it is very obvious that one particular dataset yielded higher quality results than the other 2, that being the New Vulture Dataset. Now, there can be numerous reasons for this, but the primary one could simply be the size of the dataset. Most ML models require large banks of raw data to train and build their model, and the New Vulture Dataset had significantly more of this raw data to play with (28621 data points compared to 10000 and 8607).

Eagle Subset Dataset

In the first of our datasets, the higher performing model was the Random Forrest Classifier, which achieved an F1 score of 42.91%. Although this score is far from ideal (a score of 0.9 and above is generally what is considered acceptable) it still provides us with valuable insight into how models work across different environments.

Interestingly, whilst the RNN scored a higher test accuracy (63.59%) than three other models on this dataset (KNN, ANN and AdaBoost), its F1 score was significantly lower than the rest of the tested models. Generally, this indicates poor class balance handling by the model, or misclassification in less frequent classes.

Models like KNN, SVC and Logistic regression performed with relatively mid-tier performance with F1 scores in the 37-42% range. Overall, while none of the models demonstrated anywhere near an ideal predictive capability, ensemble & tree-based methods (Gradient boosting and Random forest classifiers) generally performed better than neural networks or linear classifiers on this dataset.

New Vulture Dataset

With the highest count of data points, it's no surprise that this dataset brought exceptionally high model performance across the board. Here, once again, the Random Forest classifier model stood out as the model with the most capable predictive abilities. With an F1 score nearing perfect - 99.11% - this tree-based statistical method proved capable in this scenario.

Gradient boosting - a fellow ensemble method to the Random Forest Classifier - performed well, with an F1 score of 98.47%. The KNN model, too, was surprisingly effective, with an F1 score of 95.07%. Demonstrating the power of instance-based learning when provided with sufficient data.

As for neural networks, our ANN model proved effective with an F1 score of 86.84%. However, the RNN, while boasting the highest test accuracy out of all our tested models (99.82%), struggled to reach a high F1 score. Only obtaining 49.96%, this result could suggest overfitting of the RNN or possibly an imbalance in class distribution.

Simple models, like Logistic regression and AdaBoost, have also demonstrated high predictive capabilities. With F1 scores of 80.68% and 91.77% respectively. This suggests that the New Vulture dataset had ideal characteristics for model training, as even simple models managed to prove efficient. The main driving factor for this is likely to just be the larger size of the dataset in comparison to the other two (28,621 samples compared to 10,000 and 8607).

Traffic Roadkill Dataset

In stark contrast to the New vulture dataset, the Traffic Roadkill Dataset proved to be too challenging for all our models. The maximum obtained F1 score in this dataset was 55.87%. Once again accomplished by the Random Forest Classifier. Further proving the effectiveness of tree-based learning models in scenarios where there is a low quantity of data.

Both our neural networks - ANNs and RNNs - suffered from low generalisation power. Both had relatively acceptable test accuracy scores of 52.47% and 44.47% respectively, but had considerably lower F1 Scores of 39.37% and 27.54% respectively. This is unlikely to be attributed to simply overfitting, due to the low number of data points in this dataset.

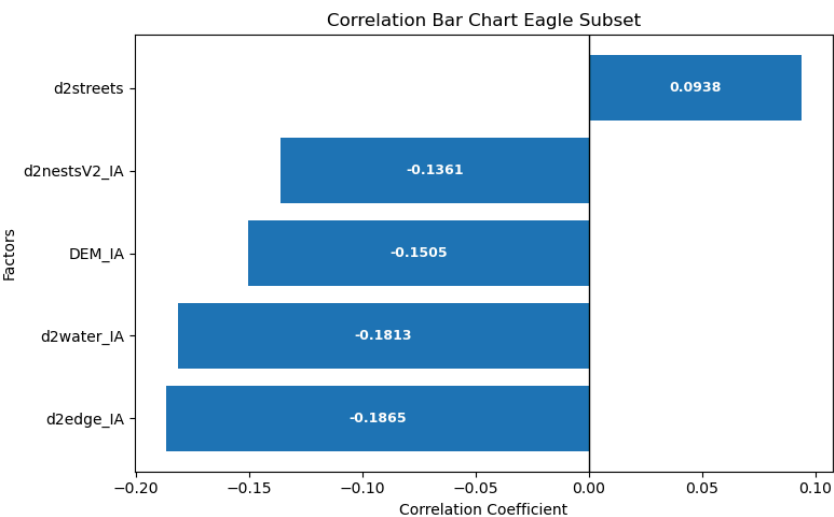
Traditional classifiers, such as Logistic Regression and SVC, struggled the most, with F1 scores of 21.86% and 32.40% respectively. Indicating a highly poor fit and therefore a dataset which is likely to be imbalanced.

Ada Boost underperformed very notably in this scenario. Scoring an F1 score of only 5.76%. This reflects a dataset which is very small and likely imbalanced.

Overall, due to the low scores across all models, the dataset is likely to either be inconsistent with its feature representation, highly imbalanced, or simply lacking in data. The latter is most likely, as the sample size was only 8607 strong in the data.

4.2. Correlation Coefficients

Alongside running several ML models, this research paper also ran several correlation tests to provide further information on what factors contribute most to WVC.



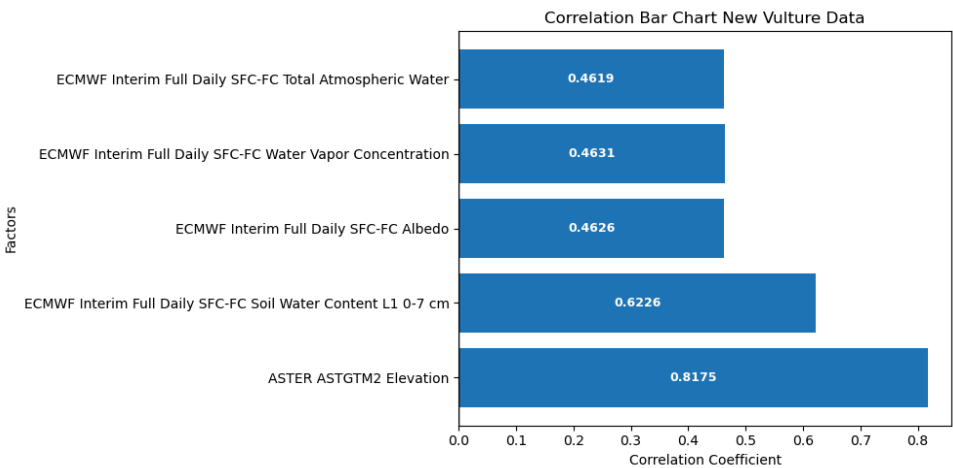
Eagle Subset Dataset

The correlation analysis conducted on the Eagle Subset Dataset provided a very subtle pattern between certain environmental factors and wildlife-infrastructure collisions between birds and wind turbines. Factors such as d2edge_IA and d2water_IA held more weight when it came down to determining whether or not a collision had occurred when

compared to other factors. However, as these scores generally fall within the weak correlation range (abs. Value of 0 - 0.3), it can be difficult to confidently assess these coefficients.

New Vulture Data

As per usual, the New Vulture Dataset excelled in providing impactful results for further analysis. The

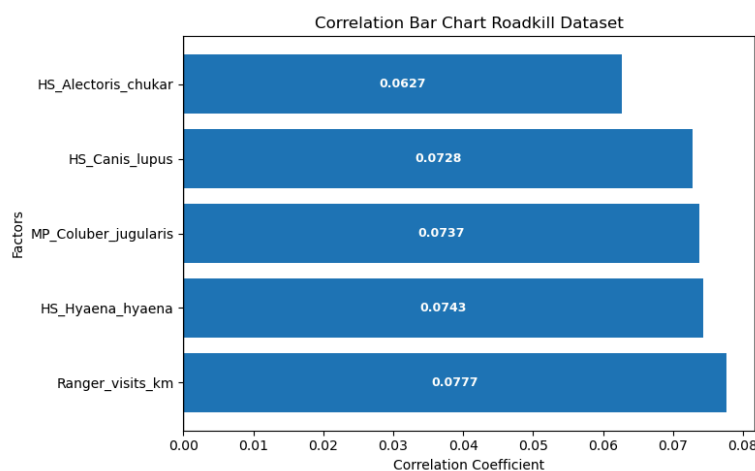


ASTER ASTGTM2 Elevation factor had an astonishingly high correlation coefficient. Scoring a value of 0.8175, this variable further reinforces its role in being a key determinant for predicting the migratory

state of vultures. Another notable variable, which had a moderately strong correlation coefficient, was that of ECMWF Interim Full Daily SFC-FC Soil Water Content L1 0-7 cm. Simply a percentage for the volume of soil moisture in surface-level soil, the positive nature of this variable's correlation coefficient suggests that vultures may be more active or more likely to migrate in regions where the upper soil layers retain higher moisture levels, potentially due to the increased availability of vegetation and prey in such environments. These findings underscore the importance of hydrological conditions and topographic features (elevation) in influencing the movement patterns of vultures.

Traffic Roadkill Dataset

The correlation analysis conducted on the Traffic Roadkill Dataset revealed several noteworthy relationships across the board between our wide range of predictor variables and the target outcome, which species was involved in the collision. Among the strongest correlations was Ranger_visits_km. A factor detailing how often a ranger visits the specific zoneage, this shows how increased ranger patrol frequency may coincide with higher recorded incidences.



Although all of these correlation coefficients fall within the ‘weak’ correlation range (abs. Value <0.3), they remain statistically relevant in ecological studies. These findings suggest that even low-magnitude correlations may carry ecological significance in the wider world when interpreted in conjunction with species-specific behaviours and their subsequent habitat preferences.

5. Conclusion and Future Work

The main aim of this study was to investigate the varying effectiveness of different machine learning models across three different wildlife-related datasets - datasets which varied in both raw size and characteristics. To do this, we applied a wide range of different models, ranging from traditional algorithms like logistic regression and KNNs to more complex ensemble methods and neural network architectures. We evaluated their predictive capabilities alongside their ability to generalise from a training dataset to determine which models are best suited for predicting Wildlife-Vehicle collisions in different data environments.

Overall, across all datasets and models tested, ensemble methods such as Random Forest and Gradient Boosting tended to deliver the most powerful predictive performances. This was especially true in the New Vulture Dataset, where both achieved near-perfect F1 scores. In contrast, simpler models such as logistic regression and KNN showed variable effectiveness depending on whether or not large quantities of well-structured data were provided. This goes to show the confident relative ability of these ensemble methods to handle both large and small datasets with complex, non-linear relationships.

As for the neural networks, ANNs generally exhibited stronger performances than RNNs. Obtaining higher F1 scores in each of the tested datasets. However, both were prone to instability or underfitting in the smaller/noisier datasets (such as the Traffic Roadkill Dataset). Moreover, RNNs occasionally produced very low F1 scores, which didn't correspond with their much higher test accuracy scores. This could be attributed to their ineffectiveness in making proper class predictions in certain scenarios. Further highlighting the importance of using a multitude of metrics, especially one which incorporates several others (i.e F1 score) - when evaluating different models.

One very clear takeaway from the workings in this study is that dataset size and quality play a hugely pivotal role in determining the success of a model. This is evident by simply glancing at the results page, whereby the New Vulture Dataset, with nearly 30,000 samples, enabled near every model to perform very well, whereas smaller datasets such as the Eagle subset (10,000 samples) and Traffic roadkill (8,607 samples) constrained the predictive power of even the more advanced models used.

However, although it is very evident that larger datasets will be needed in future cases for when more analysis is to be done on wildlife-vehicle collisions, this approach is not always feasible. In ecological & conservational domains, collecting labelled data is a highly labour-intensive task. As such, future work should explore the integration of data augmentation, transfer learning, and semi-supervised learning techniques to enhance model robustness whilst working with datasets that don't necessarily require/ have the ability to garner large quantities of labelled data. Additionally, working with feature engineering and dimensionality reduction would highly likely improve model performance across the board in smaller datasets.

In conclusion, while advanced machine learning models demonstrate great promise in reducing wildlife-vehicle collisions, future research will need to emphasise techniques which could increase the size and quality of datasets used to further the performance gains of these models. Techniques such as data augmentation and transfer learning would better support conservationists and transportation planners working in these dynamic and resource-limited environments.

REFERENCES

- [1] Federal Highway Administration Research and Technology. Chapter 2. Causes and Characteristics of Wildlife-Vehicle Collisions. Wildlife-Vehicle Collision Reduction Study: Report To Congress
<https://www.fhwa.dot.gov/publications/research/safety/08034/02.cfm>
- [2] Clevenger, A. P., Chruszcz, B. & Gunson, E. K. (2002). Spatial Patterns and Factors influencing small vertebrate fauna road-kill aggregations. Biological Conservation Volume 109, Issue 1.
- [3] Carvalho-Roel, F. C., Iannini-Custódio, E. A., Júnior, M. O. & Grilo, C. The Spatial, climatic and temporal factors influencing roadkill change according to the taxonomic level. Journal of Environmental Management volume 348.
- [4] World Wind Energy Association. Half Year Report 2024: Strong Wind Power Growth Continues in First Half 2024.
- [5] Silva, I., Crane, M., & Savini, T. (2021). The road less traveled: Addressing reproducibility and conservation priorities of wildlife-vehicle collision studies in tropical and subtropical regions. Global Ecology and Conservation Volume 27.
- [6] Owwoye, K. Forecasting Avian Migration Patterns using a Deep Bidirectional RNN Augmented with an Auxiliary Task.
- [7] Grossi, E., & Buscema, M. (2007). Introduction to artificial neural networks.
- [8] Doshi, K. (2021). Towards data science. Batch Norm Explained Visually - How it works, and why neural networks need it.
<https://towardsdatascience.com/batch-norm-explained-visually-how-it-works-and-why-neural-networks-need-it-b18919692739/>
- [9] Stryker, C. (2024, October 4). What is a recurrent neural network (RNN)? IBM.
<https://www.ibm.com/think/topics/recurrent-neural-networks>
- [10] Lee, F. (2025, May 14). What is logistic regression. IBM.
<https://www.ibm.com/think/topics/logistic-regression>
- [11] Yse, L. D. (2023, June 30). K-Nearest Neighbor (KNN) Explained.
<https://www.pinecone.io/learn/k-nearest-neighbor/>
- [12] What is random forest?. IBM.
<https://www.ibm.com/think/topics/random-forest>
- [13] (2023, December 27). What are support vector machines (SVMs)?. IBM.
<https://www.ibm.com/think/topics/support-vector-machine>
- [14] Clark, B. & Lee, F. (2025, April 07). What is gradient boosting?. IBM.
<https://www.ibm.com/think/topics/gradient-boosting>
- [15] Data Science Wizards, Understanding the AdaBoost Algorithm. Medium.
<https://medium.com/@datasciencewizards/understanding-the-adaboost-algorithm-2e9344d83d9b>